Drummond B. Fielding

Curriculum Vitae

Assistant Professor
Department of Physics, New York University, Office 1053,
726 Broadway, New York, NY 10003
drummondfielding@gmail.com dfielding14.github.io arXiv ORCID

Education

PhD 2018, MA 2014, Astrophysics, University of California, Berkeley. *Advisor: Eliot Quataert*

BS, BA 2012, Physics, Mathematics, Johns Hopkins University.

Professional Appointments

Assistant Professor, New York University, Department of Physics, Center for Cosmology and Particle Physics	2025–
Associate Research Scientist, Flatiron Institute, Center for Computational Astrophysics	2025–
Assistant Professor, Cornell University, Department of Astronomy	2024–2025
Flatiron Research Fellow, CCA	2018–2024
Visiting Scholar, Indian Institute of Science	2016
NSF & Berkeley Graduate Research Fellow, U.C., Berkeley	2012–2018

Research interests

fluid dynamics	galactic winds	stellar+BH feedback
plasma physics	circumgalactic medium	magnetic reconnection
galaxy formation	magnetic turbulence	geom. measure theory
cosmic ray transport	multiphase flows	machine learning

Grants

The Geometry of Flows, Simons Foundation, PI, 2025–2030

Pushing the Frontier of Cosmic Ray Transport in Interstellar Turbulence, DOE INCITE, PI, 2025–2027

Resolving Cosmic Ray Transport by pushing the Frontier of MHD Turbulence, DOE INCITE, PI, 2024

Illuminating the Interplay of Multiphase Galactic Winds and the CGM, NSF AAG, PI

Mocking Galactic Winds: A new approach to constraining feedback by modeling the UV spectra of galaxies, Legacy HST-AR, Co-PI, 2025–2027

Galactic Winds Unveiled: Leveraging Cloud Simulations with Radiative Transfer to Constrain Feedback, Legacy HST-AR, Co-PI, 2024–2026

Resolving Turbulent Multiphase Gas Dynamics, EURO HPC, Co-PI, 2023–2025

AthenaK: a Performance Portable Simulation Infrastructure for Computational Astrophysics, NASA Support for Open-Source Tools, Frameworks, and Libraries, Co-I, 2025–2027

DUVET: Mapping Outflows and the Inner CGM in Starbursting Disks, KCWI, Co-I, 2020–

Visiting scholar grant to work Indian Institute of Science with Prof. Prateek Sharma, APS-IUSSTF, PI, 2016

Towards an Understanding of the Origin of OVI in the Circumgalactic Medium, HST AR, Co-I, 2017

The Physics of Supernova Feedback: Global 3D Simulations of Galactic Disks, NSF XSEDE, Co-I, 2016–2018

Conduction, Convection, and Thermal Instability in Hot Halos, NSF XSEDE, Co-I, 2016–2017

Advising

Brent Tan — Postdoc — Cornell University and NYU → Polymathic

Hitesh Das — Postdoc — NYU

Alex Pendris — Graduate Student — NYU

Maëlle Magnan — Graduate Student — NYU

Nana Ama Darpaah — Masters Student — NYU

Leela Elliott — Post-bac Student — NYU

Lillion Shi — Undergraduate — NYU

Sohum Udani — Undergraduate Student — NYU

Tiegan Powell — Undergraduate Student — NYU

Yongqi Zhang — Graduate Student — Cornell University

Bodong Liu — Undergraduate Student — Cornell University

Iryna Butsky — Graduate Student — University of Washington + CCA predoctoral fellow → Hubble Fellow at Stanford

Viraj Pandya — Graduate Student — UC Santa Cruz + CCA predoctoral fellow → Hubble Fellow at Columbia

Matthew Abruzzo — Graduate Student — Columbia → Postdoc at UPitt

Anthony Chow — Graduate Student — Columbia

Zirui Chen — Undergraduate Student — Columbia → Grad Student at UCSB

Minami Roy — Graduate Student — Raman Research Institute + CCA predoctoral fellow → Postdoc at OSU

Mitali Damle — Graduate Student — University of Potsdam → Postdoc at NYU Abu Dhabi

Professional services & teaching

Founder and organizer of the *New York Area Fluid Dynamics Meeting*, a cross-disciplinary monthly meeting of local fluid dynamicists, 2020

Research Mentor, *Simons-National Society of Black Physicists Scholars Program*, 2023

Flatiron Pre-Doctoral Program graduate student mentor, 2019–2024

Referee: MNRAS, ApJ, ApJL, PRL

Proposal evaluation panelist: NSF (2020), NASA HST (2024)

Honors and awards

Outstanding Graduate Student Instructor Award	2014
NSF Graduate Research Fellowship	2014–2017
Berkeley Graduate Fellowship	2012–2014
Donald E. Kerr Award for Outstanding Physics Undergraduate	2012
The Phi Beta Kappa Society	2012

Public outreach

Founder and leader of *The Astro Club*. Monthly meetings 2020–2023. Democracy Prep Charter Middle School, Harlem, NY.

Astronomy career day, Democracy Prep Charter Middle School, Harlem, NY, 2019

Taught a month-long series of astronomy classes to 2nd and 3rd grade students, 2017. North Oakland Community Charter School, Oakland, CA

UC Berkeley Astronomy Department Public Liaison, 2015

East Bay Astronomical Society Public Lecture, 2014, Chabot Space and Science Center

The Berkeley Compass Project Summer Evening Instructor, 2013

Selected recent presentations

Invited Seminar, Diffuse Gas in Cosmic Ecosystems Science Interest Group (7/2026)
Endowed Colloquium, Niels Bohr Lecture, Niels Bohr Institute, Copenhagen (10/2024)
Seminar, Institute for Advanced Studies, Princeton (10/2024)
Colloquium, Flatiron Institute, Center for Computational Astrophysics (2/2024)
Invited Review, *New simulations for new problems in galaxy formation*, conference, Paris (12/2023)
Invited Talk, Salpeter Workshop on the Interstellar Medium, Cornell (12/2023)
Seminar, Computational Research in Boston and Beyond (CRIBB), MIT (12/2023)
Colloquium, Johns Hopkins University, Physics and Astronomy Department (11/2023)
Colloquium, Columbia University, Astronomy Department (10/2023)
Invited talk, *Interfaces and Mixing in Fluids, Plasmas, and Materials*, conference, KITP (9/2023)
Colloquium, Michigan State University, Physics and Astronomy Department (10/2023)
Invited Review, *MIST2023: Cosmic turbulence and Magnetic fields: physics of baryonic matter across time and scales*, conference (9/2023)
Invited talk, *CGM@ND*, conference (9/2023)
Invited talk, *New Views on Feedback & the Baryon Cycle in Galaxies*, conference (8/2023)
Invited talk, *Modeling multiphase astrophysical systems*, conference (5/2023)
Invited talk, *Jim's Fest/Athena++*, conference (5/2023)
Colloquium, Yale University, Astronomy Department (2/2023)
Colloquium, Cornell University, Astronomy Department (1/2023)
Invited Astrophysics Seminar, UIUC (11/2022)
Joint Astrophysics Colloquium, Pitt-CMU (10/2022)
Invited talk, *What Matter(s) Around Galaxies*, conference (9/2022)
Invited talk, *Ringberg Computational Galaxy Formation 2022*, conference (4/2022)
Invited talk, *Wolfe Symposium*, conference (3/2022)
Invited Seminar, Theoretical AstroPhysics Including Relativity & Cosmology, Caltech (12/2021)
Astronomy Department seminar, Racah Institute of Physics, Hebrew University of Jerusalem (12/2021)
Invited Talk, *Baltimore Wind Workshop*, Johns Hopkins University (8/2021)
Astrophysics Colloquium, UCSD-SDSU (4/2021)
Keynote speaker, KITP workshop *Fundamentals of Gaseous Halos* (1/2021)
Invited talk, KITP conference *On the Origin, Nature, and Mixing of Multiphase Gas in Astrophysics* (10/2020)
Gaggle Seminar, Carnegie Institution for Science (8/2020)
Invited talk, CLEARNESS conference IAP (6/2020)
CGI Colloquium, U.C. Santa Cruz (4/2020)
Contributed Talk, Universality of Turbulence Conference, Flatiron Institute, NY (12/2019)
SFIR Seminar, Princeton University (11/2019)
Astro Seminar, NYU CCPP, New York, NY (10/2019)
Invited Talk, CGM conference, Berlin, Germany (10/2019)
Invited seminar, Turbulence workshop, Aspen Center for Physics (6/2019)
Invited Talk, *athena++* conference, Las Vegas, NV (3/2019)
Invited Talk, CGM conference, Northwestern, Chicago, IL (8/2018)
Invited Talk, CGM/DLA conference, Big Sur, CA (3/2018)
TAPIR Seminar, Caltech, Pasadena, CA (11/2017)
Astronomy Colloquium, Raman Research Institute (12/2016)
Astronomy Department Colloquium, Indian Institute of Science (10/2016)

Publications Summary— *ADS search — Google Scholar — ORCID*

total publication count: 86 — citations: 4162 — h-index: 37 (*8 Nov 2026*)
* indicates $\geq$ 100 citations — ‡ graduate student-led — § undergraduate-led — † postdoc-led